

SHALKAR, Kazakhstan

WMO: 356330

Lat: 47.850N

Lon: 59.617E

Elev: 577

StdP: 14.39

Time Zone: 5.00 (E05)

Period: 84-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.4	-15.1	-24.4	1.5	-18.5	-20.2	1.9	-14.4	37.5	11.8	34.5	9.1	6.8	270	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	96.4	65.3	93.4	64.3	90.1	63.3	67.8	86.8	66.5	86.8	65.2	85.4	10.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	86.2	72.2	60.0	78.9	72.0	57.5	72.0	71.8	32.4	86.0	31.4	87.2	30.4	85.5	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.1	26.7	23.1	DB	-25.0	102.3	2.6	2.3	-26.9	104.0	-28.5	105.3	-30.0	106.6	-31.9	108.2	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	9.9	6.0	18.4	46.5	60.1	73.0	76.0	72.2	59.9	43.3	25.3	14.4
	DBStd	27.03	12.76	11.63	12.59	10.72	7.96	7.49	6.00	6.46	8.08	8.74	11.28	11.79
	HDD50	5760	1244	1231	979	185	15	0	0	0	16	244	742	1104
	HDD65	9210	1709	1651	1444	558	188	18	2	16	190	673	1192	1569
	CDD50	2945	0	0	0	81	329	690	807	689	312	37	0	0
	CDD65	916	0	0	0	3	37	257	343	240	36	0	0	0
	CDH74	11800	0	0	0	50	582	3480	4347	2782	539	20	0	0
	CDH80	5331	0	0	0	7	146	1701	2135	1199	140	3	0	0
Wind	WSAvg	10.4	10.8	11.6	12.2	11.6	10.7	10.2	9.9	9.1	9.7	9.5	10.2	9.5
Precipitation	PrecAvg	7.3	0.6	0.5	0.6	0.7	0.5	0.6	0.6	0.5	0.5	0.7	0.7	0.6
	PrecMax	16.6	1.5	1.7	1.8	6.8	2.2	3.1	2.0	1.8	2.9	2.1	2.1	1.4
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0
	PrecStd	2.6	0.4	0.4	0.4	1.2	0.5	0.6	0.6	0.6	0.6	0.5	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	33.7	54.3	79.6	88.9	101.1	101.9	97.1	87.6	75.9	49.7	36.3
		MCWB	32.9	32.2	45.7	57.7	60.0	66.4	69.1	64.5	58.7	54.8	44.9	33.5
	2%	DB	32.6	32.0	44.8	73.8	83.5	96.9	96.5	93.3	84.0	69.8	43.8	34.0
		MCWB	31.8	31.1	39.6	54.7	58.6	64.9	65.9	63.6	58.3	51.6	38.0	32.6
	5%	DB	30.4	28.5	39.2	68.8	79.9	92.7	93.7	89.6	80.3	63.7	41.4	32.2
		MCWB	29.3	27.4	35.7	52.3	57.6	63.8	65.0	63.1	57.3	48.3	37.2	30.7
	10%	DB	27.8	24.3	35.0	64.1	76.6	89.1	90.6	86.5	76.2	58.7	38.7	30.4
		MCWB	26.8	23.3	32.7	49.5	56.3	62.4	64.2	62.2	55.8	46.5	34.9	28.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.2	32.7	46.1	59.3	63.5	68.7	74.5	67.5	64.3	56.5	46.1	34.3
		MCDB	33.5	33.3	54.1	73.3	78.0	88.5	93.7	86.1	76.5	74.3	48.2	35.0
	2%	WB	31.9	31.1	39.9	56.4	61.2	67.0	68.9	66.3	61.6	52.5	40.3	32.7
		MCDB	32.6	32.0	44.1	71.8	77.4	89.3	87.0	84.7	74.7	65.8	42.1	33.7
	5%	WB	29.5	27.6	35.7	53.3	59.4	65.6	67.1	65.1	59.3	49.9	37.8	31.0
		MCDB	30.4	28.6	38.9	66.6	75.3	89.1	86.2	83.4	76.0	61.5	40.3	32.2
	10%	WB	26.9	23.5	32.9	50.5	57.9	64.0	65.8	63.8	57.2	47.1	35.0	28.7
		MCDB	27.6	24.3	34.8	62.7	73.2	85.8	85.8	81.6	74.7	57.2	37.8	30.2
Mean Daily Temperature Range	5% DB	MDBR	14.7	16.8	16.0	20.6	23.6	25.1	23.7	23.9	25.2	22.5	14.3	13.7
		MCDBR	13.4	12.7	16.7	26.1	29.2	29.5	27.3	27.8	31.2	31.0	17.9	12.5
		MCWBR	13.1	12.3	12.1	13.1	12.1	10.0	9.0	10.4	14.1	17.0	14.0	10.9
	5% WB	MCDBR	13.4	12.4	15.8	25.7	25.0	26.1	23.5	24.0	26.7	28.0	15.4	11.2
		MCWBR	13.5	12.2	11.7	13.9	11.6	10.5	9.2	9.8	13.1	16.5	13.3	10.3
Clear-Sky Solar Irradiance	taub	0.308	0.334	0.402	0.438	0.436	0.430	0.419	0.417	0.365	0.360	0.340	0.317	
	taud	2.100	2.022	2.018	2.037	2.118	2.197	2.247	2.233	2.330	2.257	2.221	2.133	
	Ebn at Noon	231	252	252	257	263	265	266	261	266	244	220	210	
	Edh at Noon	28	38	46	50	48	44	42	41	33	30	25	24	
All-Sky Solar Radiation	RadAvg	467	824	1171	1586	2036	2206	2134	1888	1460	875	448	311	
	RadStd	81	116	111	155	139	134	101	102	72	82	49	53	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page